

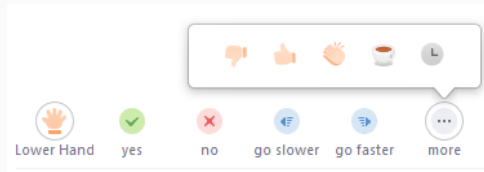
Algorithms for Lattice Fermions User Workshop 2024

15-19th Jul 2024

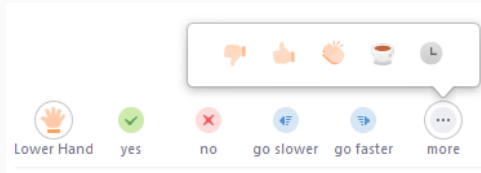
Welcome



1. Please **raise** your hands:



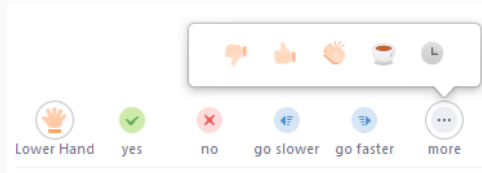
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2. Keep yourself **muted** during presentations in order to avoid background noise.

Organizational remarks

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3. Got any questions/comments? Please use the Zoom **chat**.
4. If you encounter any issues, please let us know in the chat, in Discord (<https://discord.gg/VppWWEPMHa>), or send an email to alf@physik.uni-wuerzburg.de

1. Organizational remarks

2. The ALF Project

ALF Features

ALF Team

ALF Timeline & Sponsors

3. Workshop Program

Algorithms for Lattice Fermions: general code for auxiliary-field quantum Monte Carlo simulations

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General Hamiltonian

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- squares of single-body operators
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- near-optimal single-core performance
- restart facilities
- DHF5 data files

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- extensive documentation
- Predefined Structures
- scripts and templates

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Advanced Features

- parallel tempering
- ground state projective QMC
- global Monte Carlo updates
- continuous fields
- Langevin dynamics updates
- stochastic Maximum Entropy
- symmetric Trotter decomposition
- Dimer-Dimer correlations
- Cotunneling (Kondo)
- Rényi Entropy

ALF Collaboration

ALF Team members:



Fakher Assaad



Florian Goth



Anika Götz



Johannes Hofmann



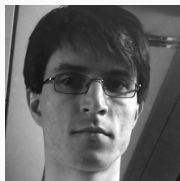
Emilie Huffman



Zihong Liu



Francesco Parisen
Toldin



Jefferson S.E.
Portela



Jonas Schwab



Maksim Ulybyshev

A short history of ALF:

2016 ALF 0.5

2017 ALF 1.0

M. Bercx, F. Goth, J. S. Hofmann, and F. F. Assaad, SciPost Phys. **3** (2017), 013

2020 ALF 2.0, pyALF

2022 ALF 2.4; Association ALF e.V. is founded

New reference paper: ALF Collaboration, SciPost Phys. Codebases **1** (2022)

2023 ALF 2.5



SFB1170
ToCoTronics



Leibniz-Rechenzentrum
der Bayerischen Akademie der Wissenschaften



GCS
Gauss Centre for Supercomputing



Center of excellence – complexity and
topology in quantum matter



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- **Keynote talks**:
 - “Hybrid Monte Carlo methods”, **Peter Lunts** (Harvard University)
 - “Constrained Path Quantum Monte Carlo”, **Shiwei Zhang** (CCQ, Flatiron Institute, Simons Foundation)
 - “Lefschetz thimbles” **Masafumi Fukuma** (Kyoto University)

Workshop Program

Time (GMT+1)	Friday 12.07	Monday 15.07	Tuesday 16.07	Wednesday 17.07	Thursday 18.07	Friday 19.07	
08:30							
09:00					Contributed talks (1+2)	HMC (1)	HMC (2)
10:00	ALF/pyALF installation	Welcome Event	Coffee break	Coffee break	Coffee break	Coffee break	
11:00		Fermion Monte Carlo (1)	Tutorial	Tutorial	CPMC(1)	CPMC(2)	
12:00							
13:00		Lunch	Lunch	Lunch	Lunch	Lunch	
14:00		Tutorial intro	ALF Basics: Observables	ALF Basics: Hopping	Thimbles(1)	Thimbles(2)	
15:00		ALF overview + Lattices	Fermion Monte Carlo (2)	Fermion Monte Carlo (3)			
16:00		Coffee break	Coffee break	Coffee break	Coffee break	Coffee break	
17:00		ALF/pyALF installation	Tutorial	Tutorial Advanced	ALF Basics: Model Building	Advanced pyALF	Workgroup/ Presentaion
18:00	ALF Basics: Interactions			Workgroup	Workgroup/ Discussion	Closing talk	
19:00					Conference Dinner		
20:00							